

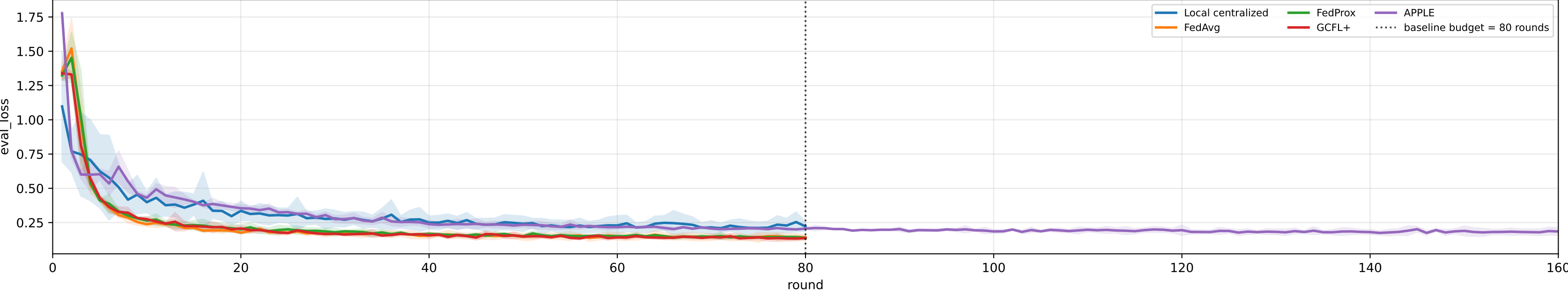
Controlled task-specialization level: High | aggregated / pooled performance over 5 clients
| model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
family=specialized_cycle3 | n_clients=1 | cycle2=visible 1.3%±0.0% / true 6.4% | cycle3=visible 1.3%±0.0% / true 6.3% | cycle4=visible 1.3%±0.0% / true 6.6% |
family=specialized_cycle4 | n_clients=1 | cycle2=visible 1.3%±0.0% / true 6.4% | cycle3=visible 1.3%±0.0% / true 6.3% | cycle4=visible 1.3%±0.0% / true 6.6% |
family=specialized_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
family=specialized_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 |
family_centroid_jsd_mean=0.162

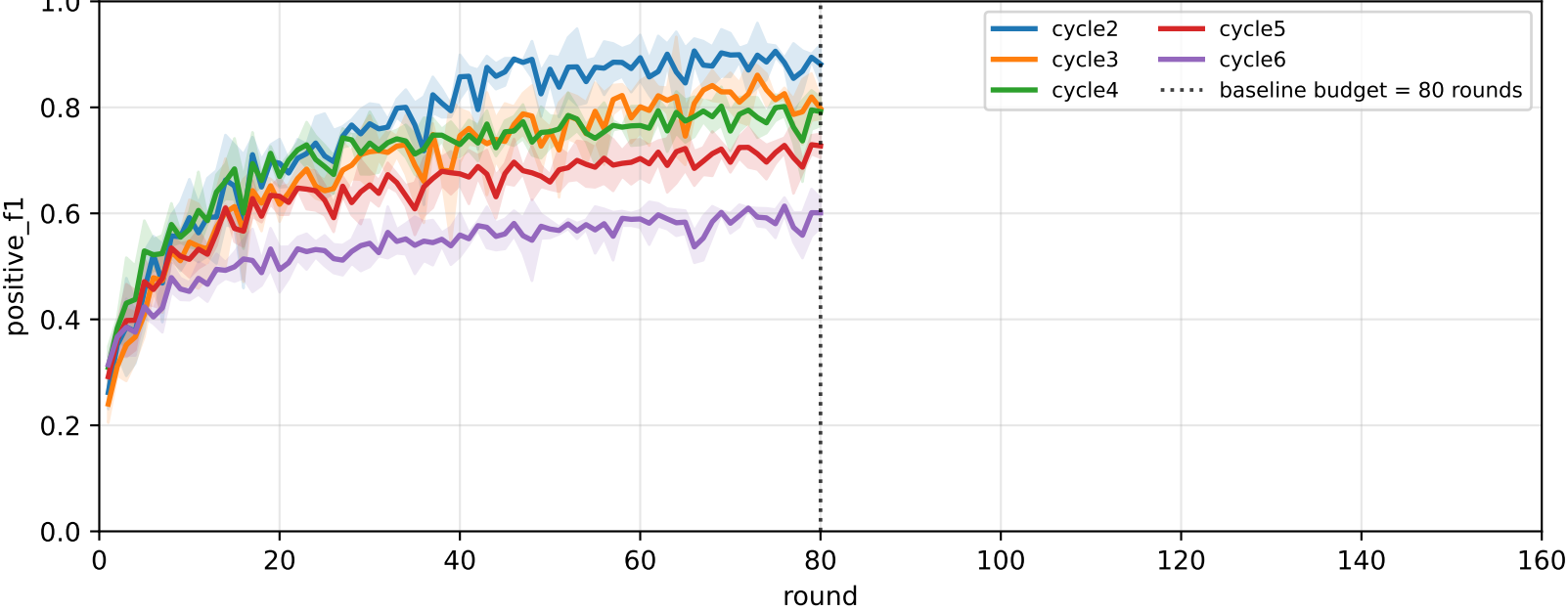
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	74.7% ± 1.4%	78.7% ± 1.0%	91.5% ± 2.6%	89.1% ± 0.9%	80.9% ± 1.9%	68.6% ± 3.9%	58.8% ± 1.7%
FedAvg	76.8% ± 1.8%	80.5% ± 1.9%	95.1% ± 1.3%	92.3% ± 3.1%	81.6% ± 1.4%	74.6% ± 3.1%	59.0% ± 1.0%
FedProx	76.0% ± 1.4%	80.0% ± 1.0%	94.9% ± 0.7%	92.6% ± 2.2%	83.5% ± 1.6%	70.0% ± 3.6%	58.9% ± 2.2%
GCFL+	77.8% ± 1.7%	80.5% ± 1.3%	95.5% ± 0.6%	90.2% ± 0.9%	78.7% ± 2.0%	72.8% ± 0.8%	65.0% ± 3.6%
APPLE	76.0% ± 3.2%	79.0% ± 2.8%	92.5% ± 1.3%	87.2% ± 2.2%	78.7% ± 3.0%	73.5% ± 3.2%	61.3% ± 4.5%

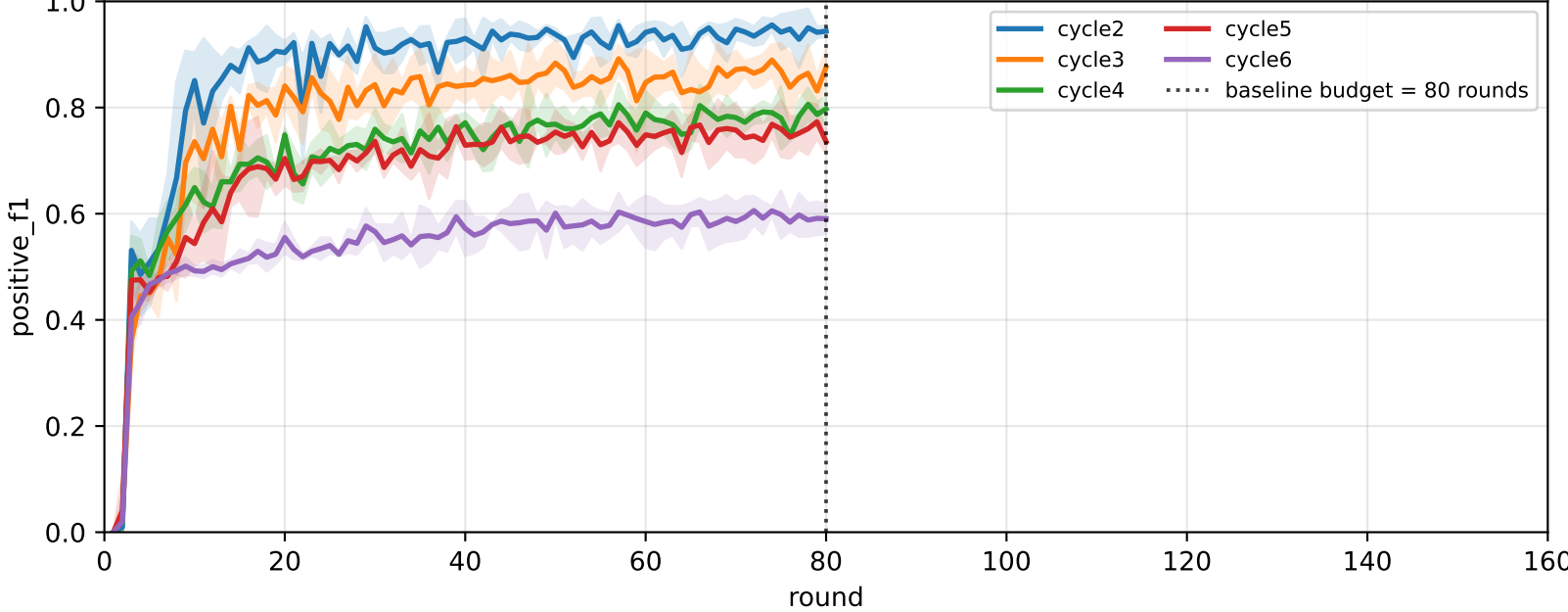
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



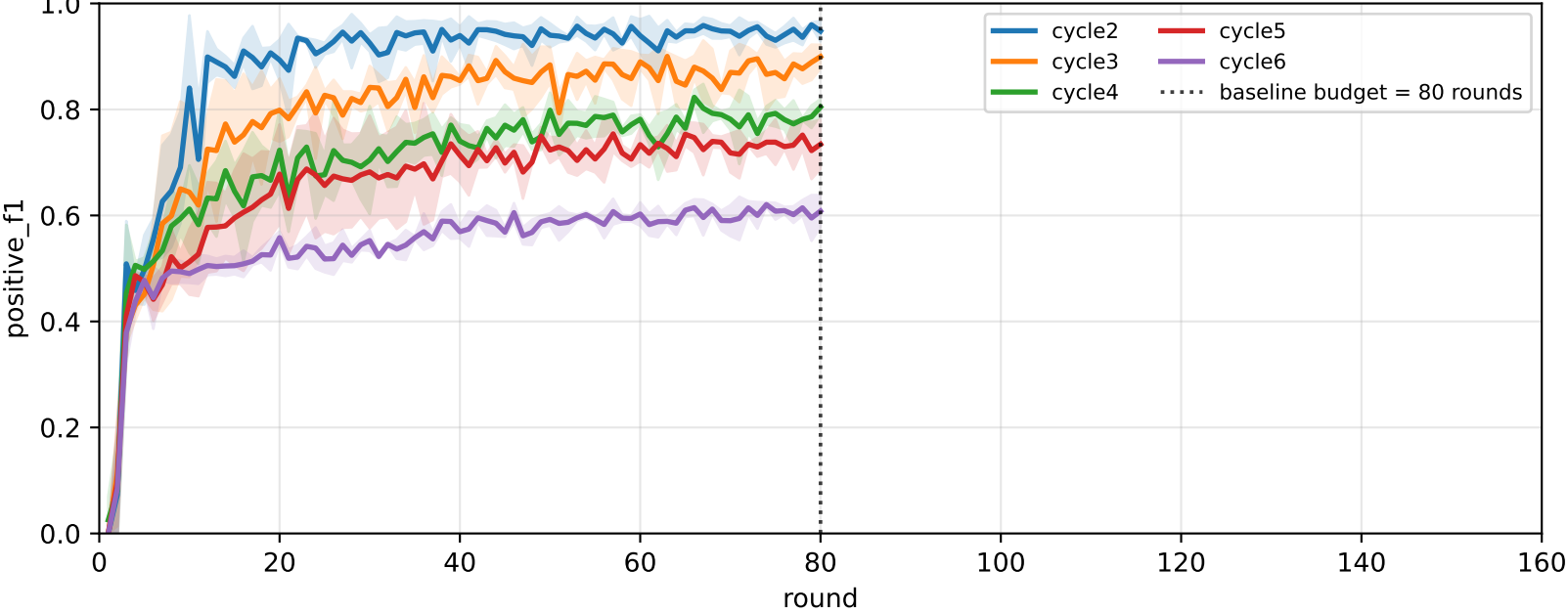
Pooled local centralized positive F1



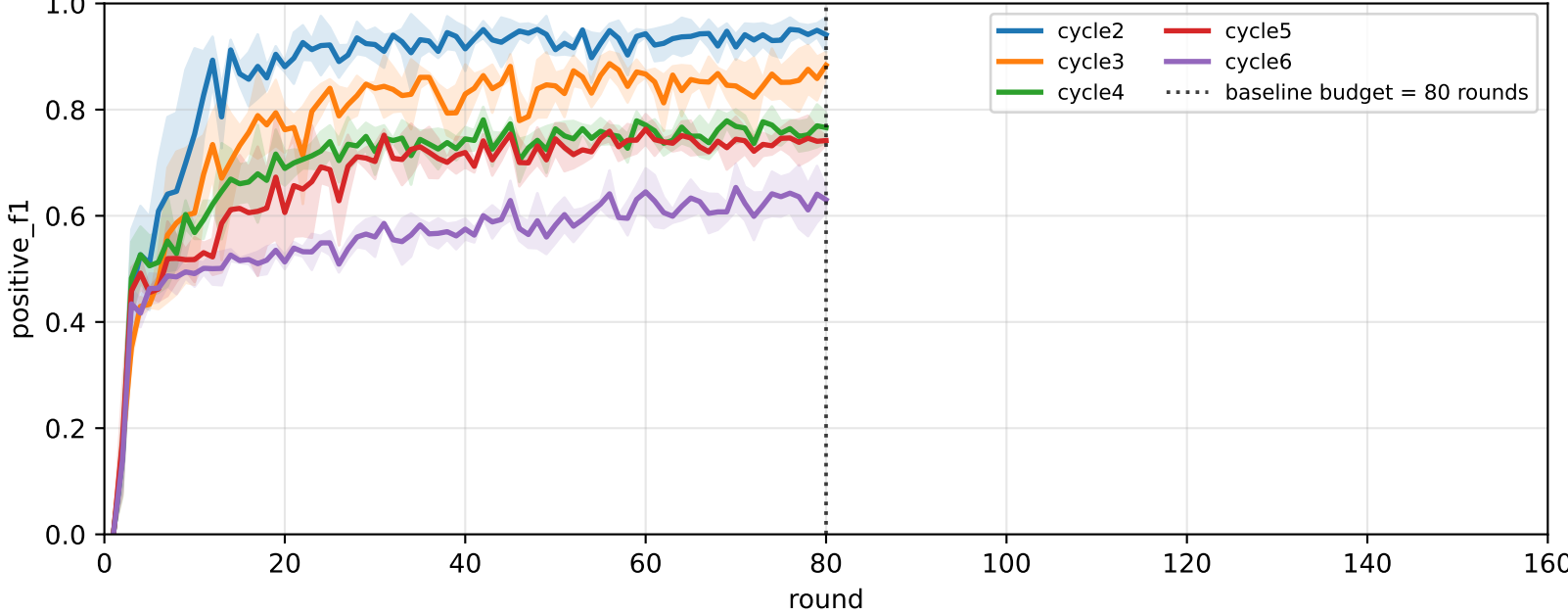
Pooled FedAvg positive F1



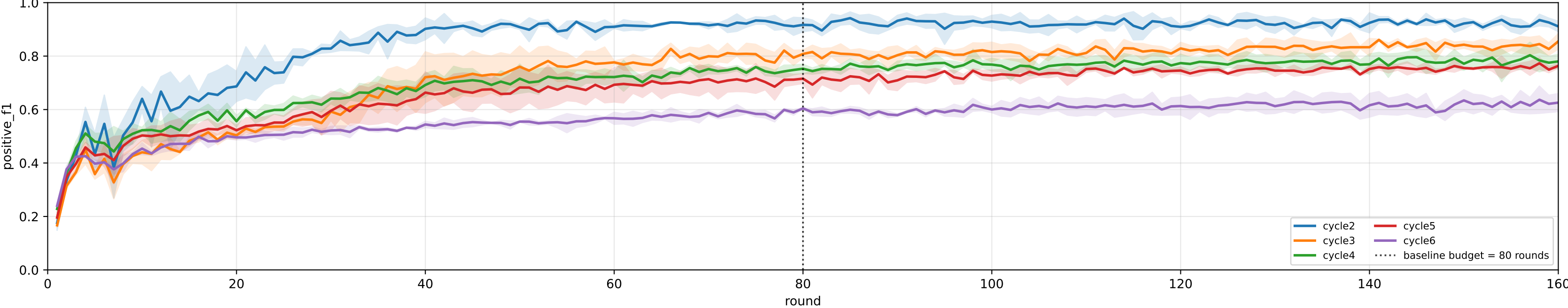
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



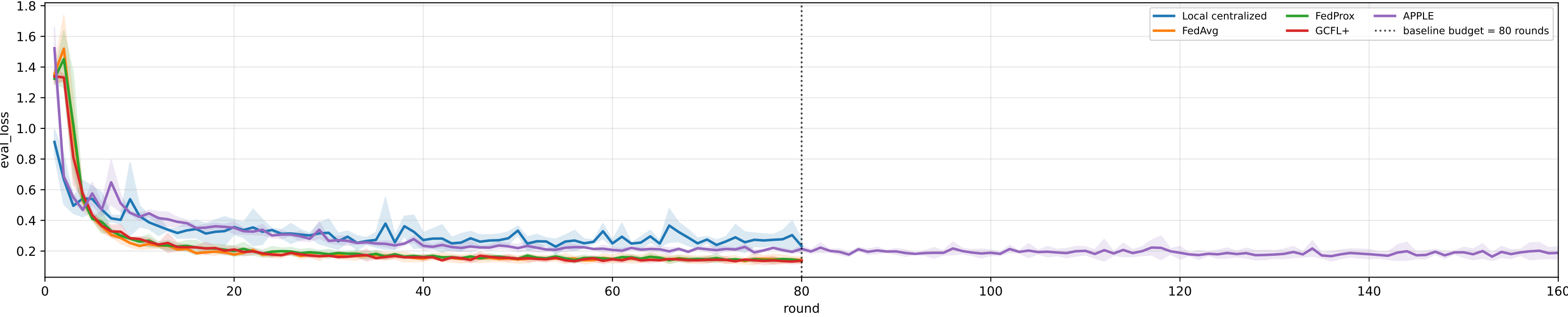
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle2 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle2 | n_clients=1 | cycle2=visible 5.8%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 |
family_centroid_jsd_mean=0.162

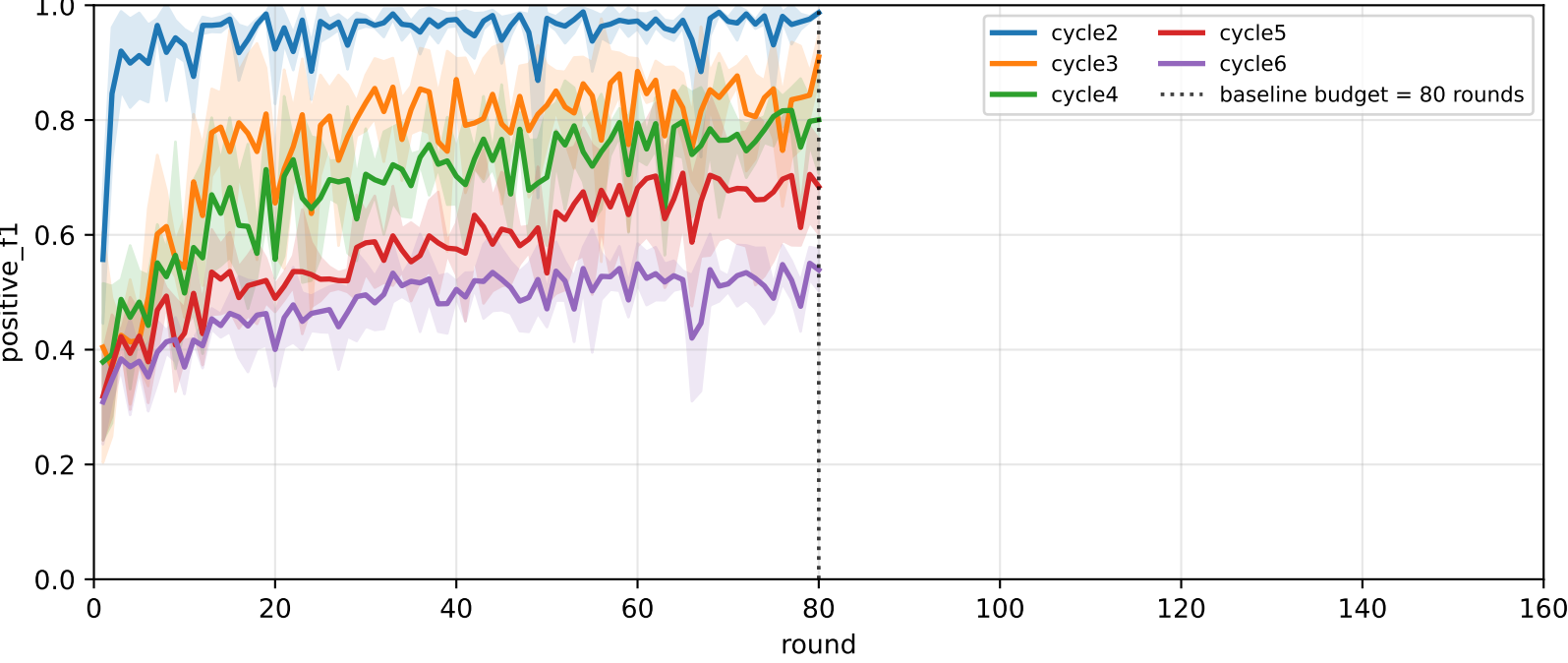
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	71.8% ± 2.2%	76.9% ± 0.8%	98.1% ± 0.8%	90.4% ± 2.4%	76.9% ± 4.5%	65.2% ± 5.3%	53.7% ± 4.1%
FedAvg	77.0% ± 1.8%	80.7% ± 1.8%	95.6% ± 0.6%	92.2% ± 3.4%	81.6% ± 1.2%	74.7% ± 2.9%	59.3% ± 1.1%
FedProx	76.0% ± 1.5%	80.1% ± 1.3%	95.9% ± 1.6%	92.6% ± 2.2%	83.0% ± 2.5%	70.1% ± 3.3%	58.8% ± 2.0%
GCFL+	78.3% ± 1.7%	80.9% ± 1.3%	95.6% ± 0.7%	90.4% ± 1.2%	79.1% ± 2.1%	73.9% ± 0.4%	65.6% ± 3.8%
APPLE	77.6% ± 3.8%	81.1% ± 3.3%	96.9% ± 2.5%	90.9% ± 3.1%	82.0% ± 3.9%	75.6% ± 4.4%	60.0% ± 5.1%

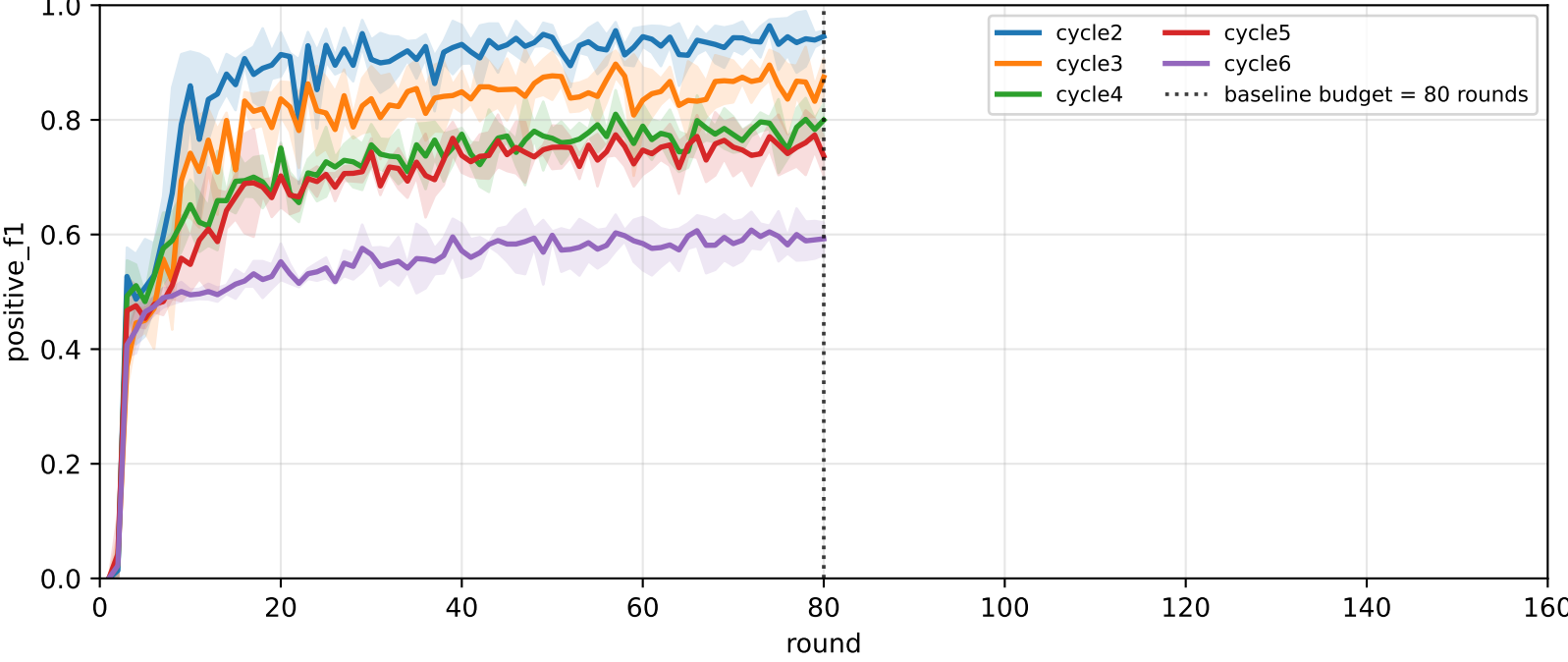
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



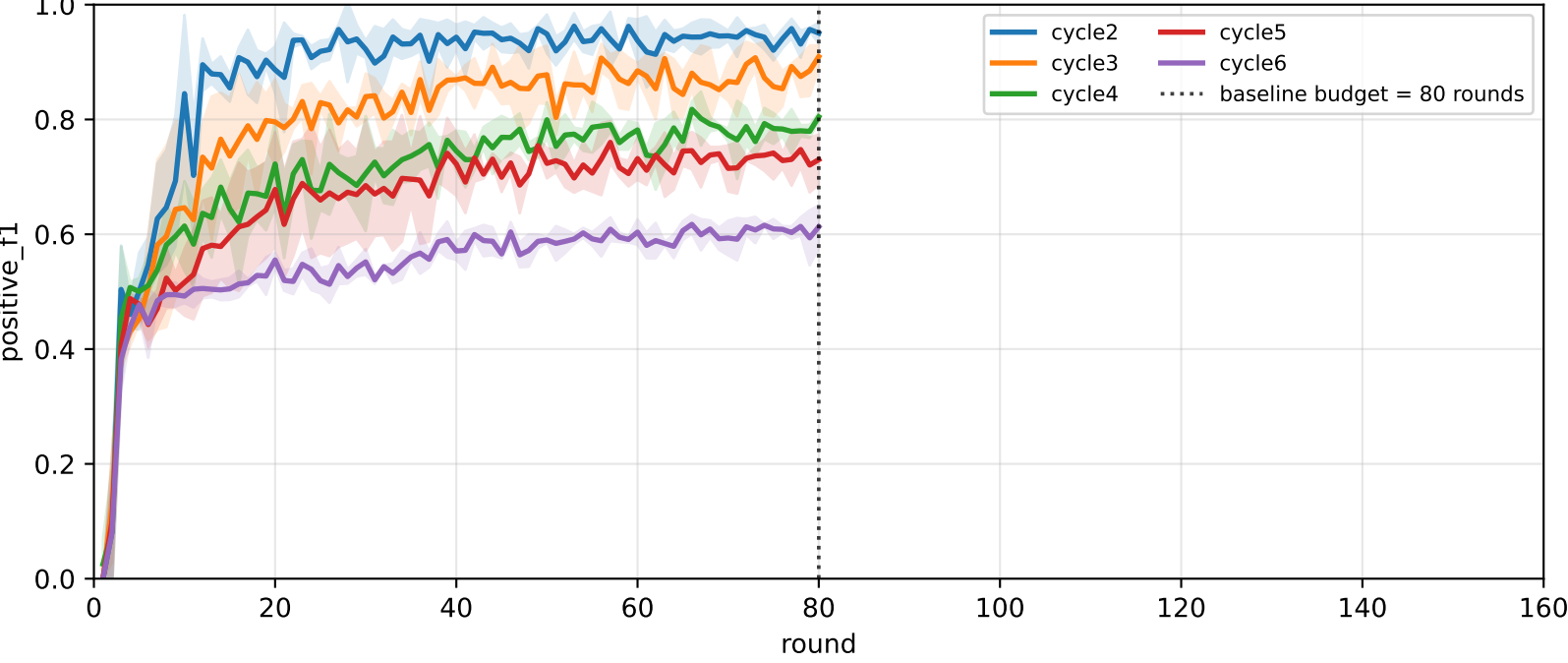
Pooled local centralized positive F1



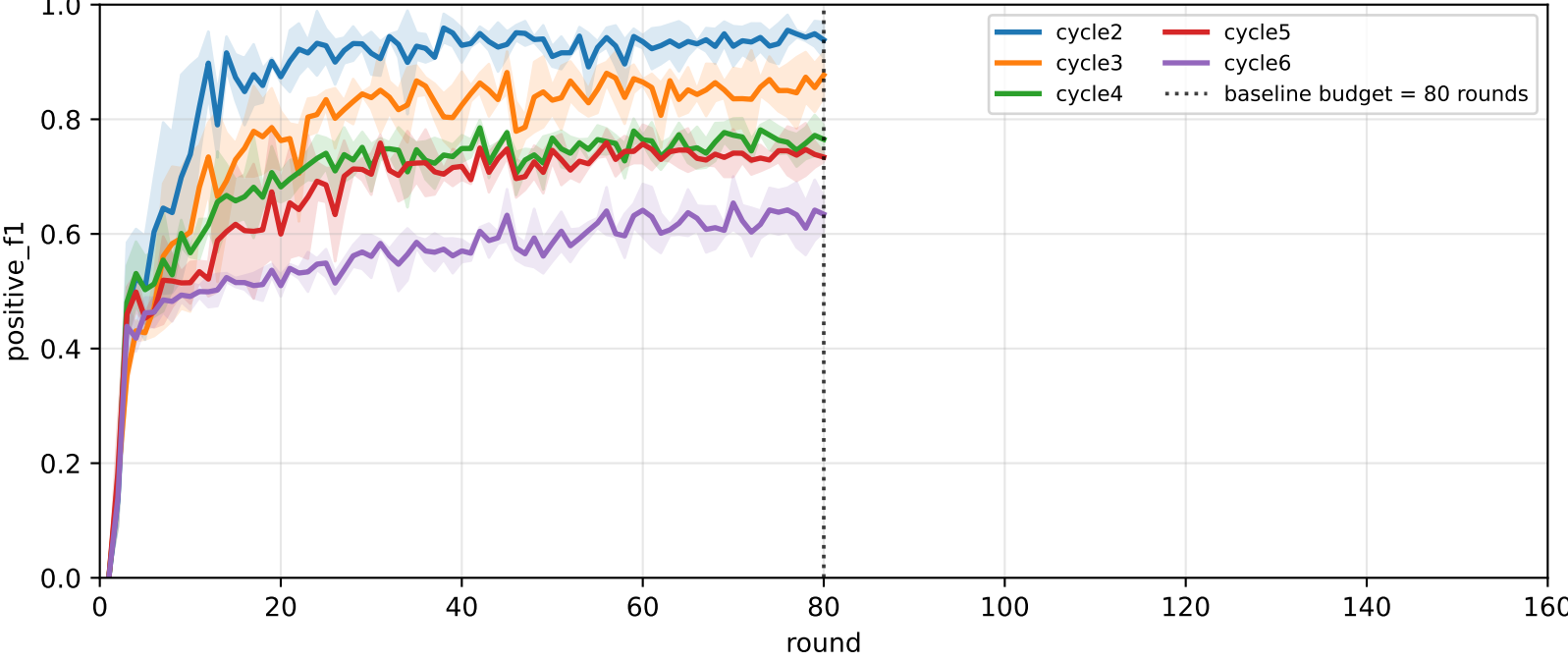
Pooled FedAvg positive F1



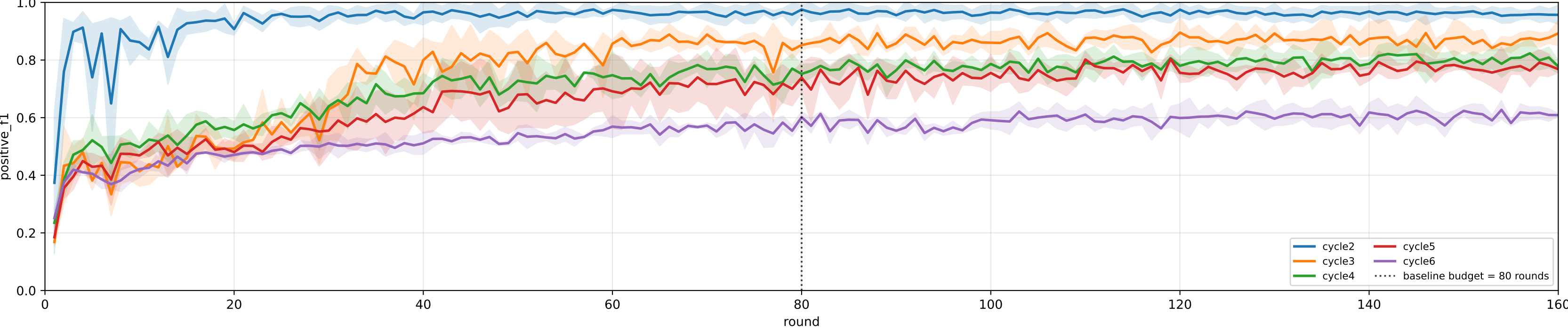
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



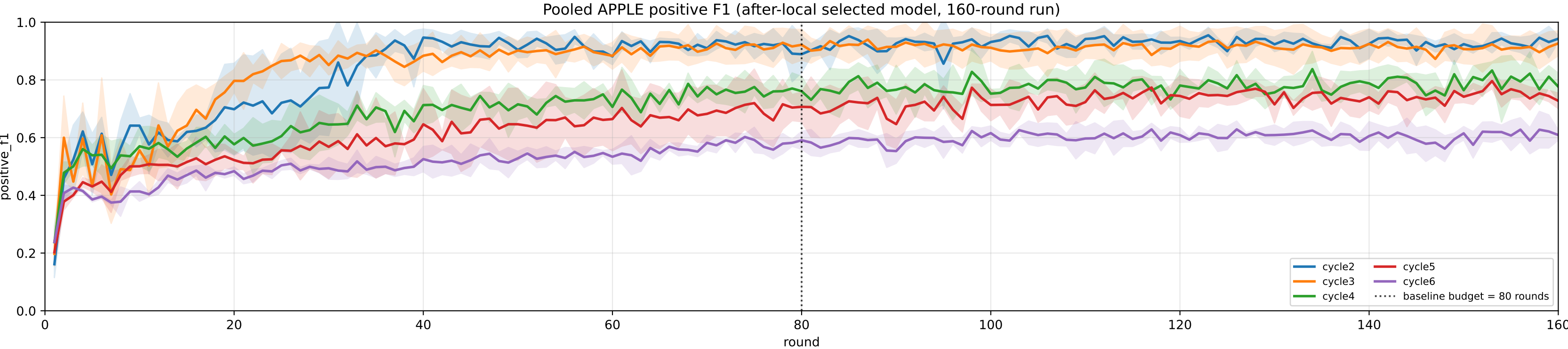
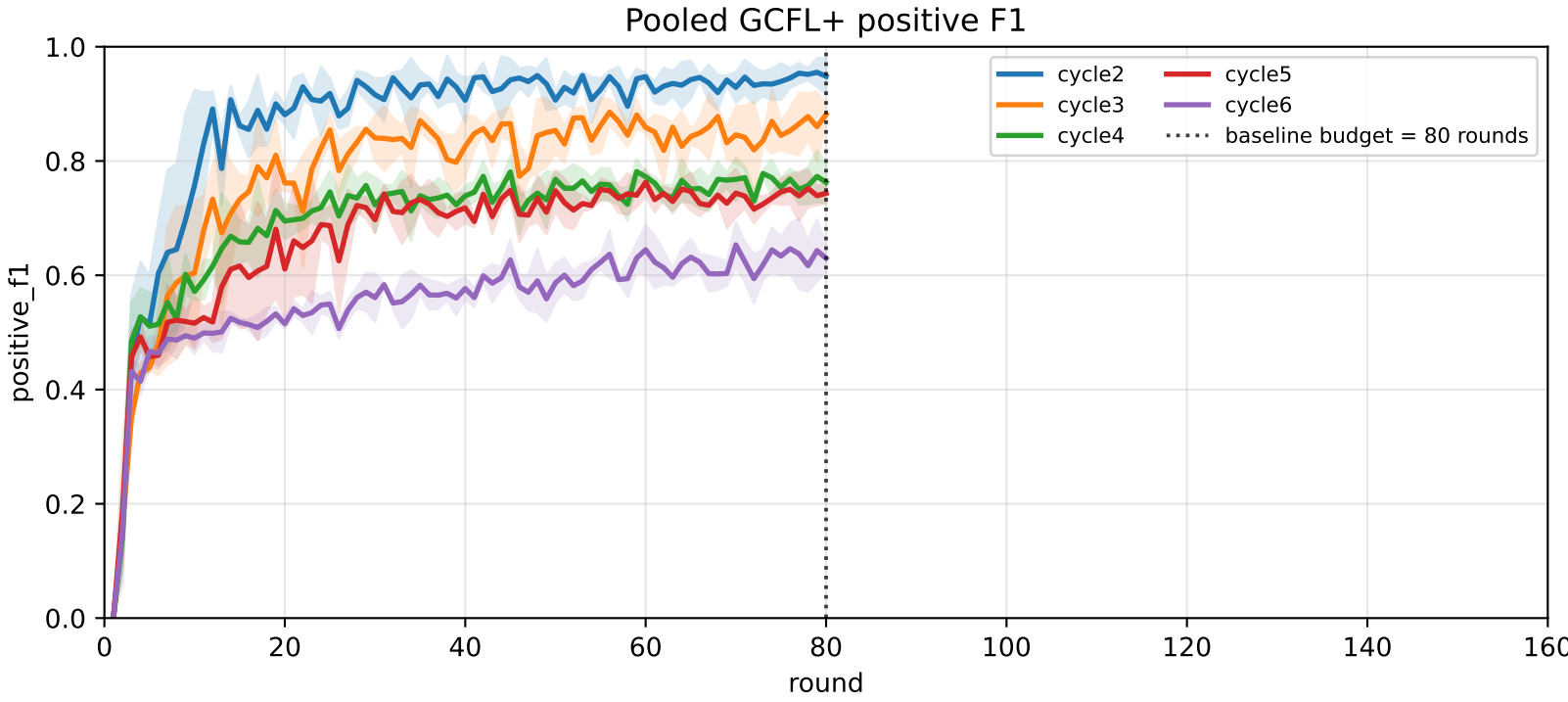
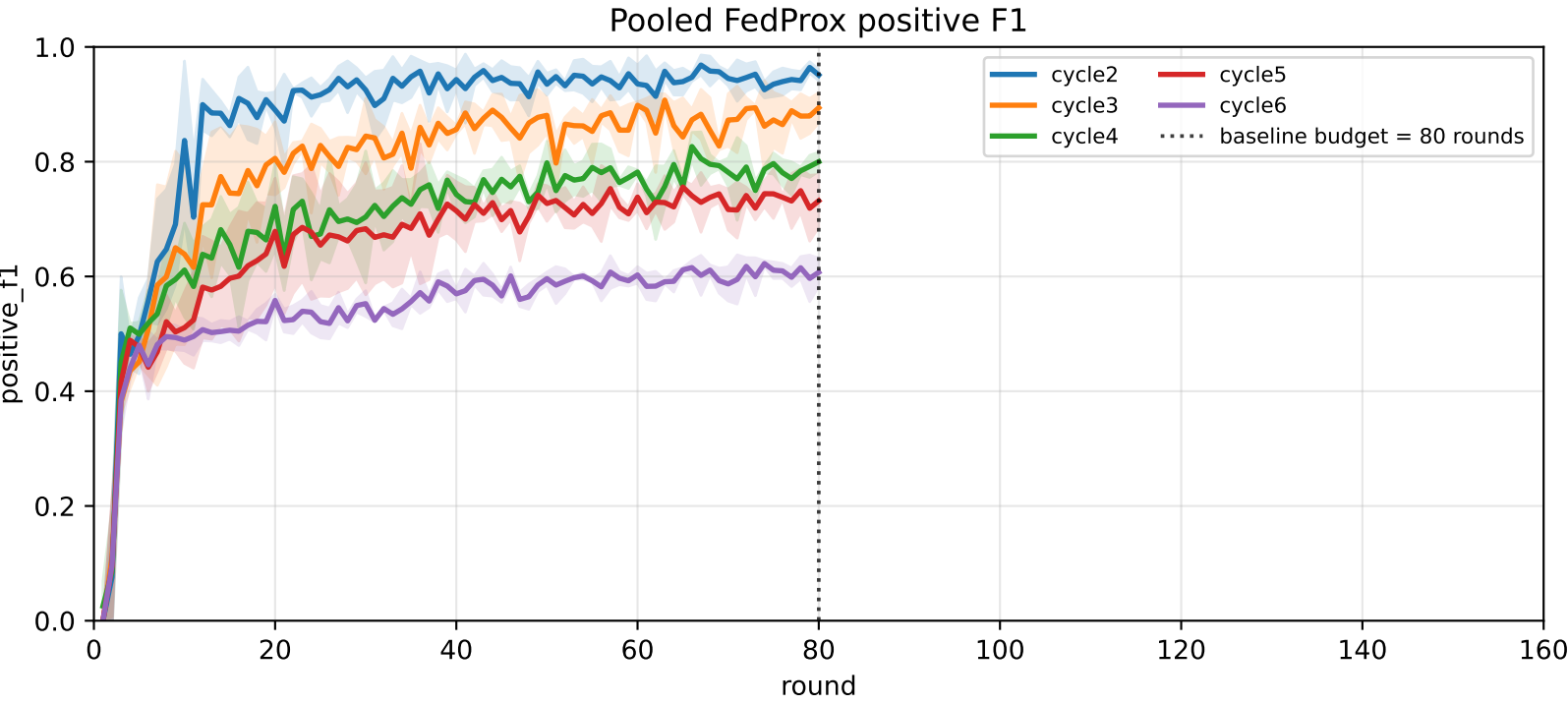
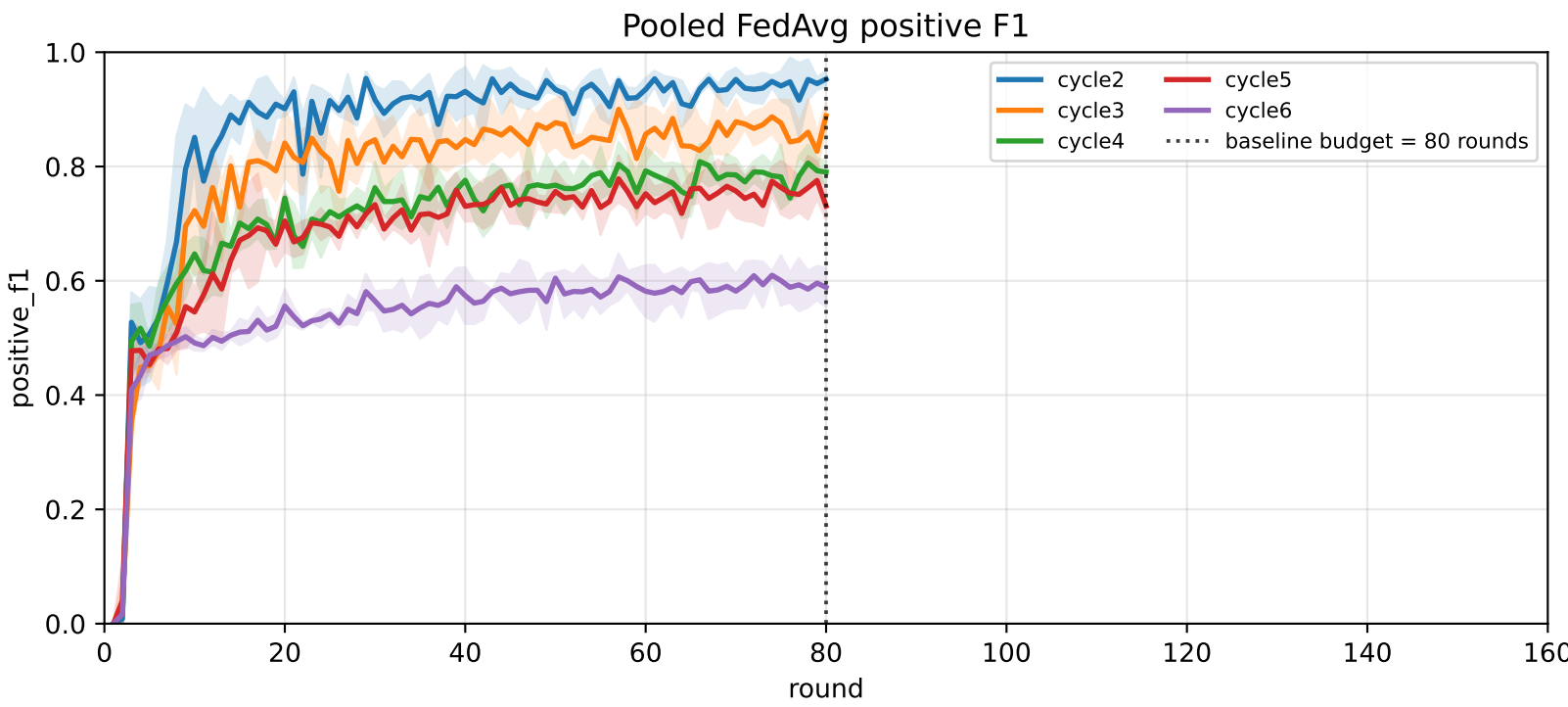
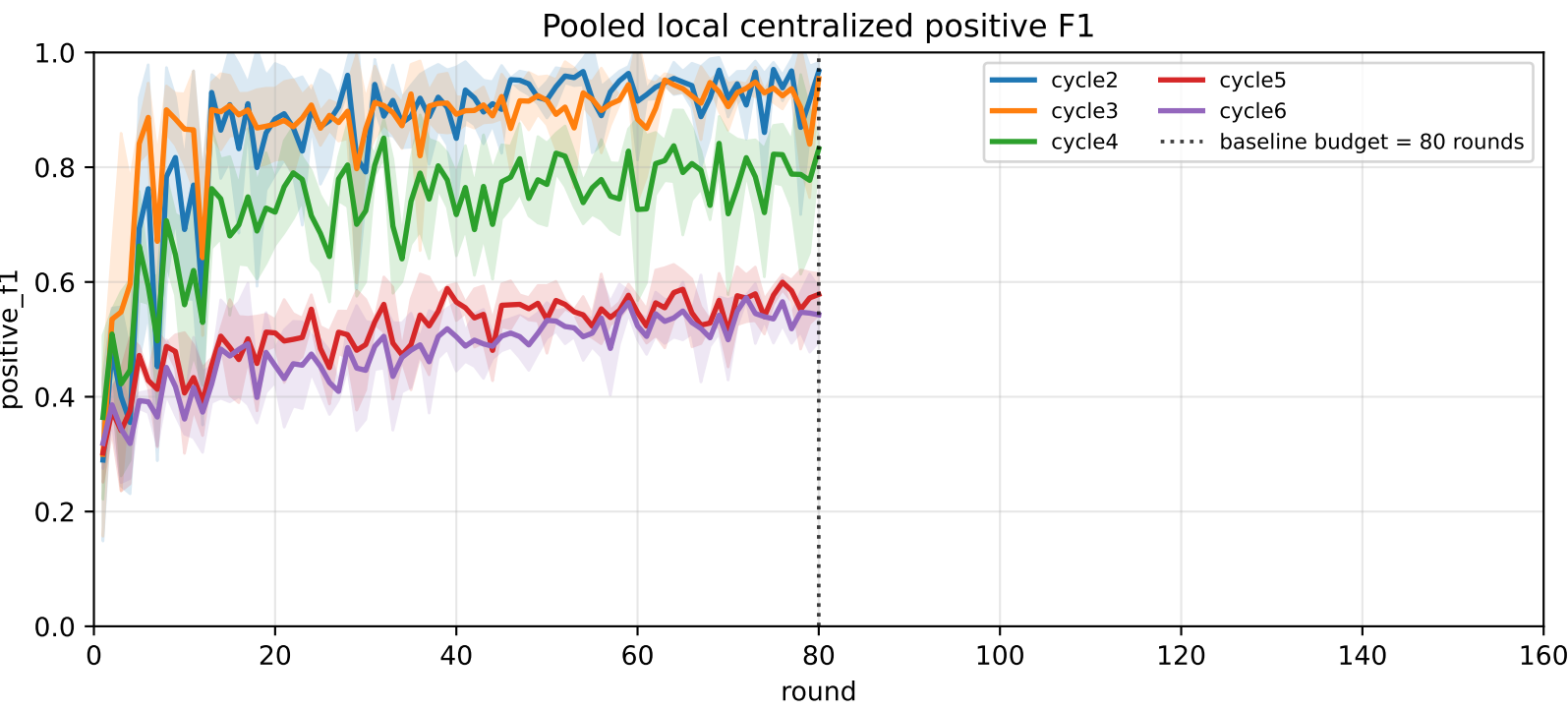
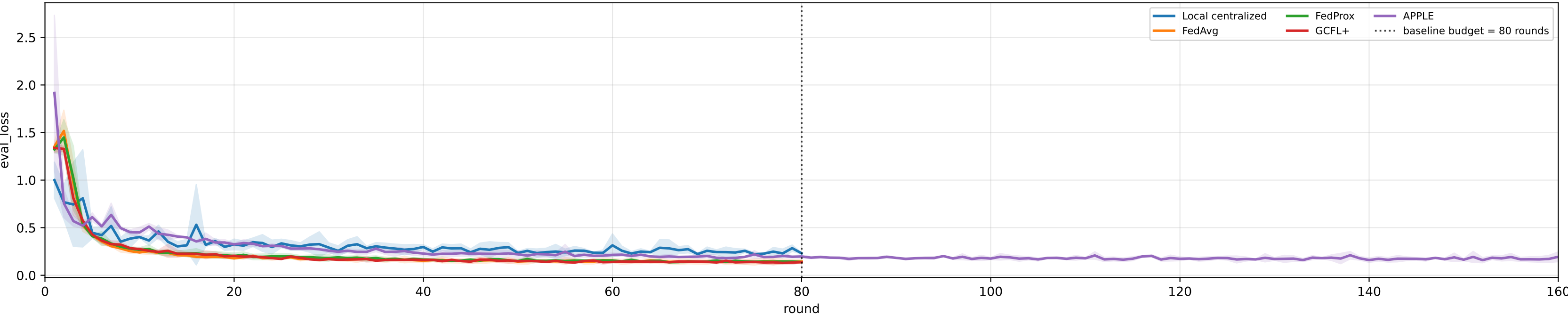
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle3 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle3 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 6.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 |
family_centroid_jsd_mean=0.162

Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	69.6% ± 3.7%	76.6% ± 3.0%	96.4% ± 1.1%	94.7% ± 1.9%	86.7% ± 5.4%	52.7% ± 5.2%	52.7% ± 4.1%
FedAvg	76.9% ± 1.3%	80.7% ± 1.4%	94.9% ± 1.8%	93.0% ± 2.3%	82.0% ± 0.5%	74.4% ± 2.1%	59.0% ± 1.0%
FedProx	76.2% ± 1.7%	80.2% ± 1.1%	95.1% ± 0.8%	93.3% ± 2.4%	83.6% ± 2.2%	70.0% ± 4.8%	59.2% ± 2.6%
GCFL+	78.0% ± 2.0%	80.7% ± 1.5%	95.6% ± 0.3%	90.8% ± 1.6%	79.5% ± 1.5%	72.8% ± 1.4%	65.0% ± 3.7%
APPLE	75.6% ± 2.5%	78.9% ± 2.4%	91.1% ± 2.1%	93.4% ± 3.7%	78.7% ± 3.9%	72.2% ± 1.0%	59.2% ± 2.7%

Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



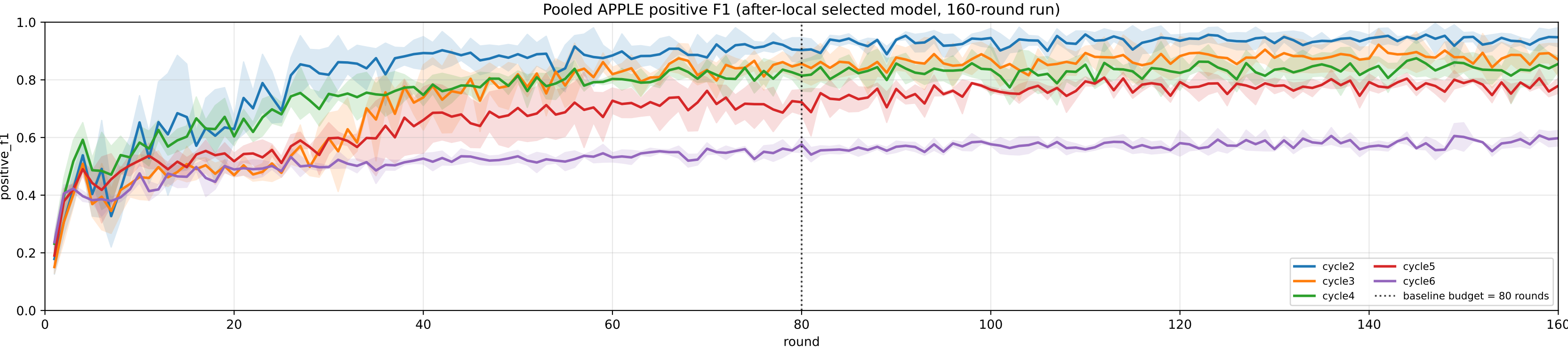
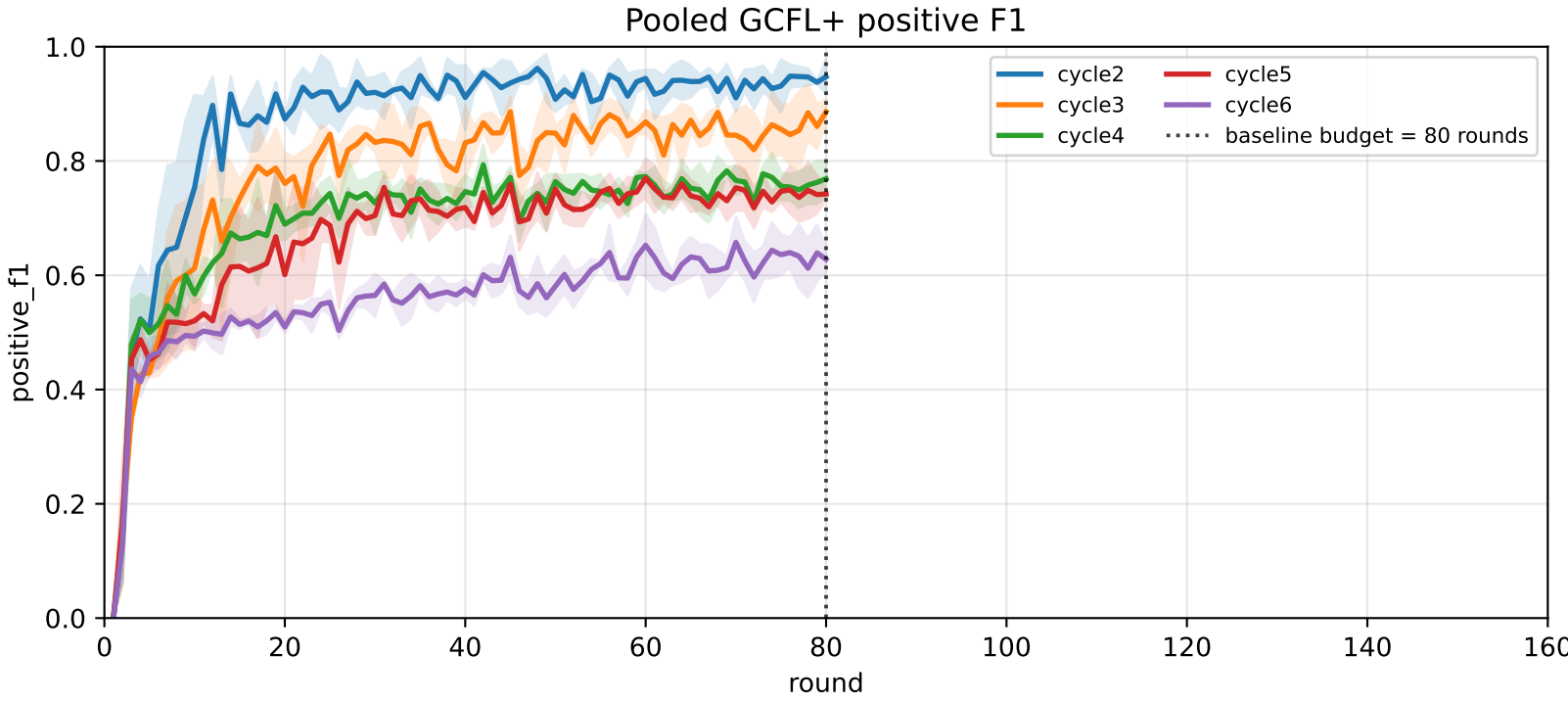
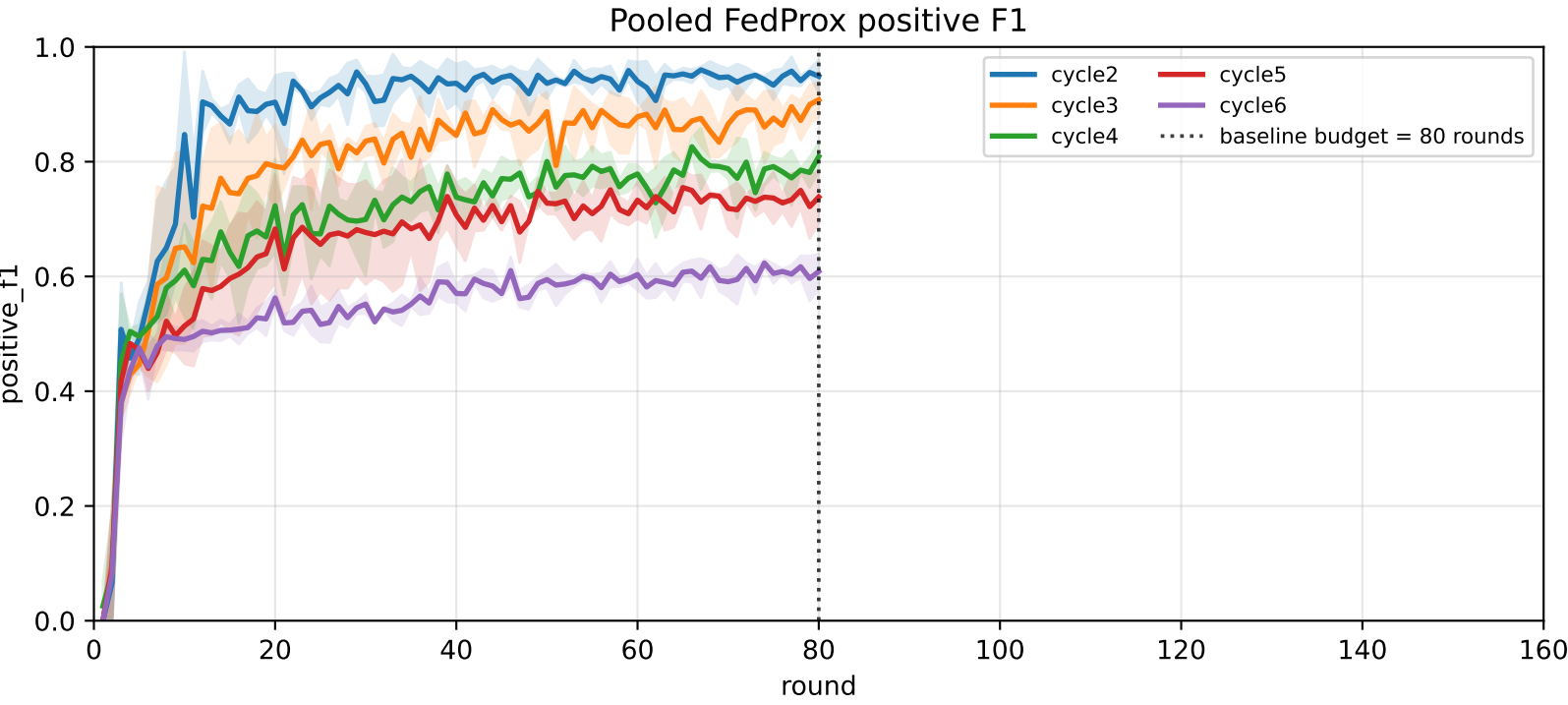
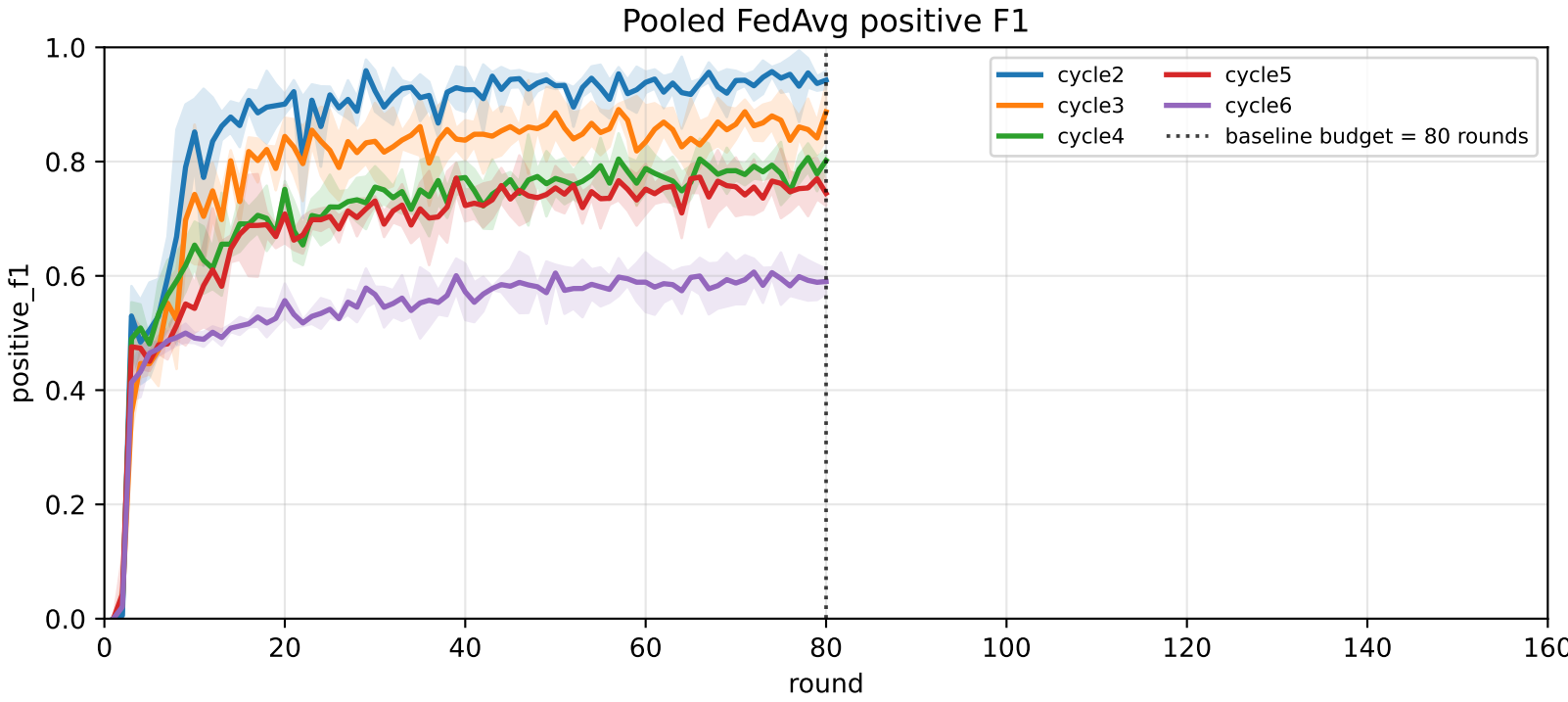
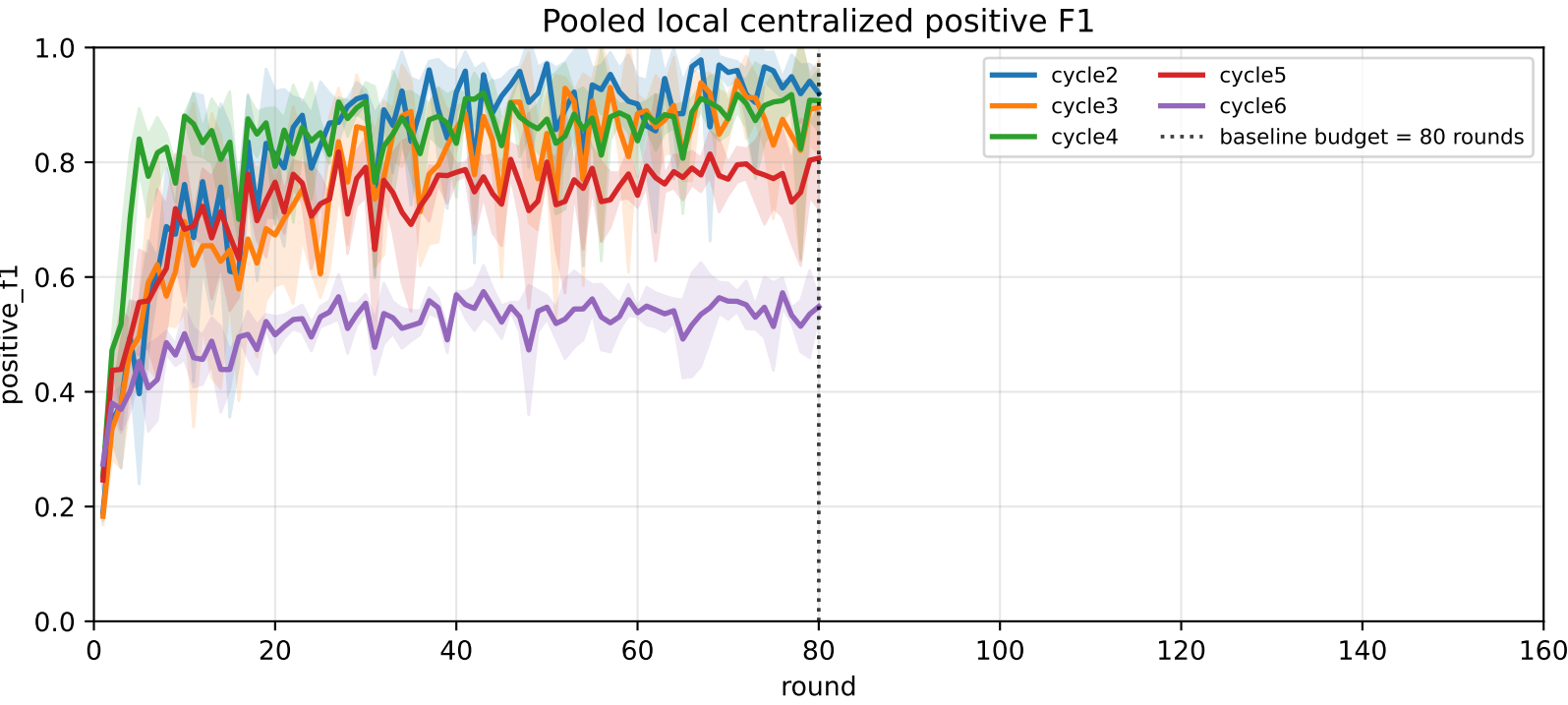
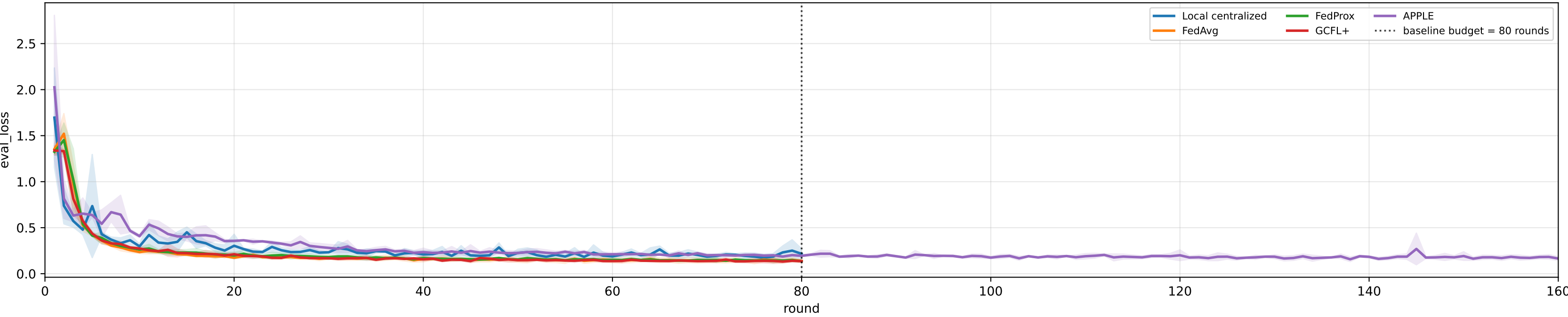
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle4 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle4 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 6.6%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3% |
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 |
family_centroid_jsd_mean=0.162

Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	78.0% ± 1.1%	82.8% ± 1.1%	95.4% ± 2.6%	95.6% ± 1.0%	91.0% ± 2.8%	76.7% ± 2.4%	55.2% ± 0.7%
FedAvg	76.7% ± 1.9%	80.5% ± 1.9%	95.2% ± 0.1%	92.3% ± 3.6%	81.3% ± 1.8%	74.7% ± 3.4%	59.0% ± 1.3%
FedProx	75.8% ± 1.4%	79.7% ± 0.9%	95.2% ± 1.3%	92.1% ± 2.4%	83.3% ± 1.0%	69.5% ± 3.4%	58.6% ± 2.2%
GCFL+	77.4% ± 2.1%	80.0% ± 1.5%	95.1% ± 0.6%	89.1% ± 0.4%	78.6% ± 2.3%	71.9% ± 1.5%	65.2% ± 4.2%
APPLE	76.8% ± 3.8%	80.4% ± 3.7%	94.4% ± 2.7%	89.1% ± 3.5%	85.2% ± 5.6%	76.2% ± 4.6%	57.1% ± 2.7%

Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



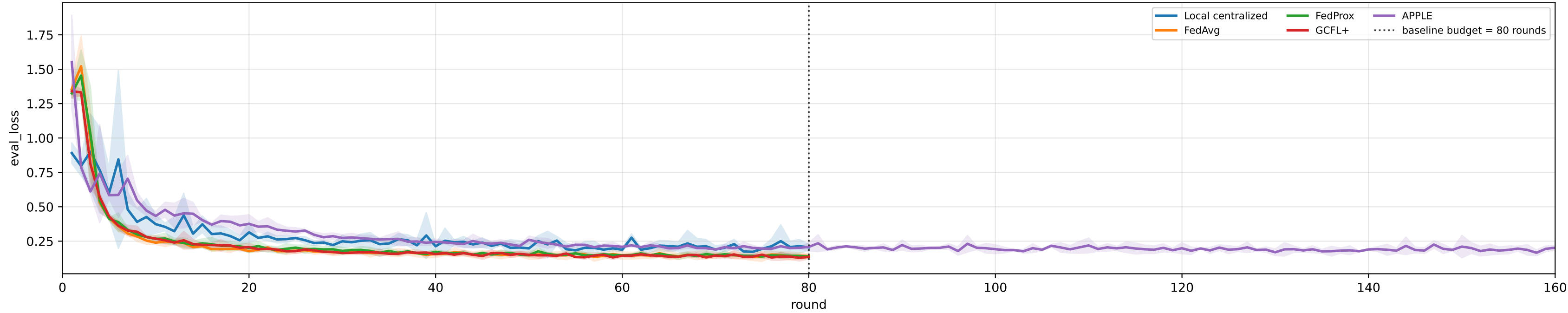
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle5 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle5 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 6.4%±0.0% / true 6.4% | cycle6=visible 1.3%±0.0% / true 6.3%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 |
family_centroid_jsd_mean=0.162

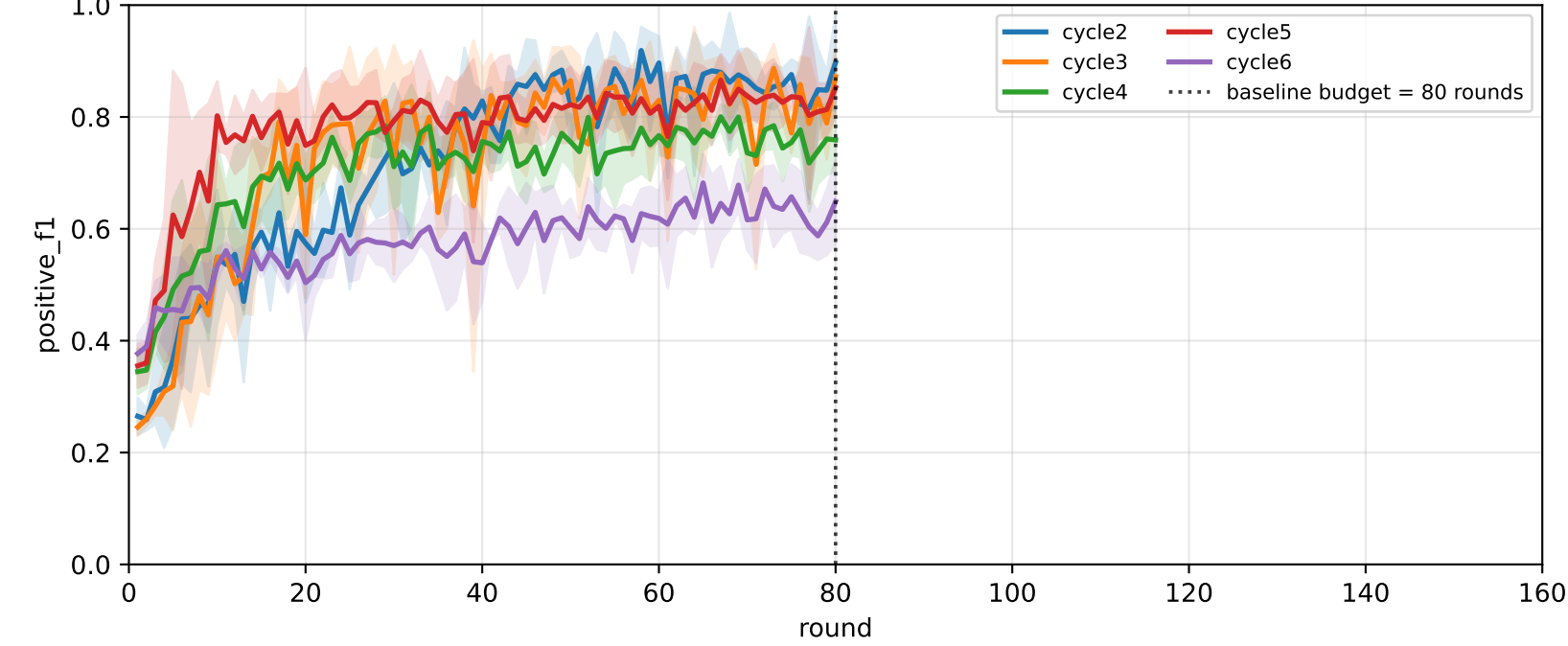
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	75.8% ± 0.9%	78.3% ± 1.5%	87.0% ± 1.3%	87.4% ± 3.9%	77.0% ± 3.6%	79.2% ± 5.6%	60.7% ± 3.5%
FedAvg	76.5% ± 1.9%	80.2% ± 2.0%	94.7% ± 0.9%	91.3% ± 3.3%	81.3% ± 1.5%	74.4% ± 3.6%	59.0% ± 1.0%
FedProx	75.7% ± 1.5%	79.5% ± 1.3%	93.2% ± 1.4%	91.8% ± 2.7%	84.0% ± 1.6%	69.5% ± 3.3%	58.9% ± 2.3%
GCFL+	77.8% ± 1.2%	80.5% ± 0.7%	95.5% ± 0.8%	90.2% ± 0.8%	78.5% ± 2.2%	73.7% ± 1.4%	64.4% ± 3.2%
APPLE	76.3% ± 3.5%	79.5% ± 3.0%	93.0% ± 0.9%	87.9% ± 4.1%	80.0% ± 6.4%	77.6% ± 2.2%	58.8% ± 4.2%

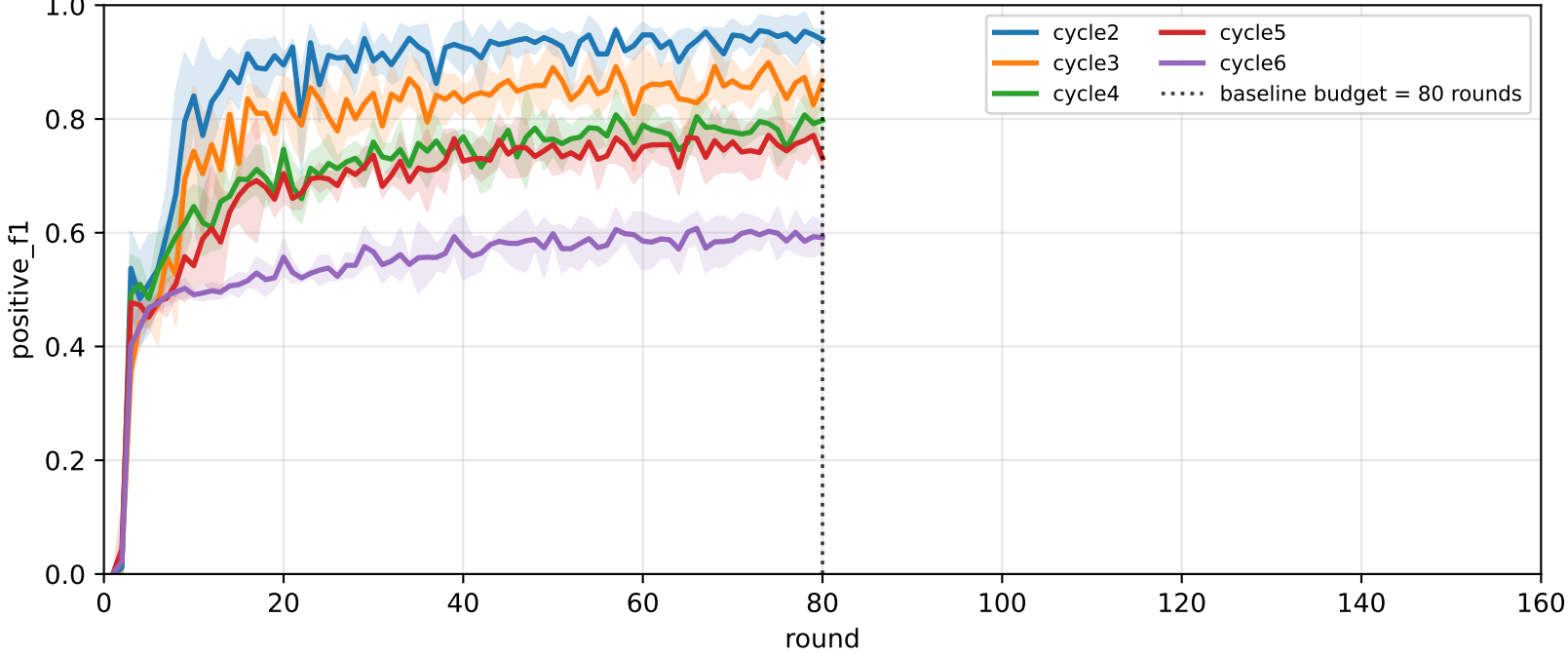
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



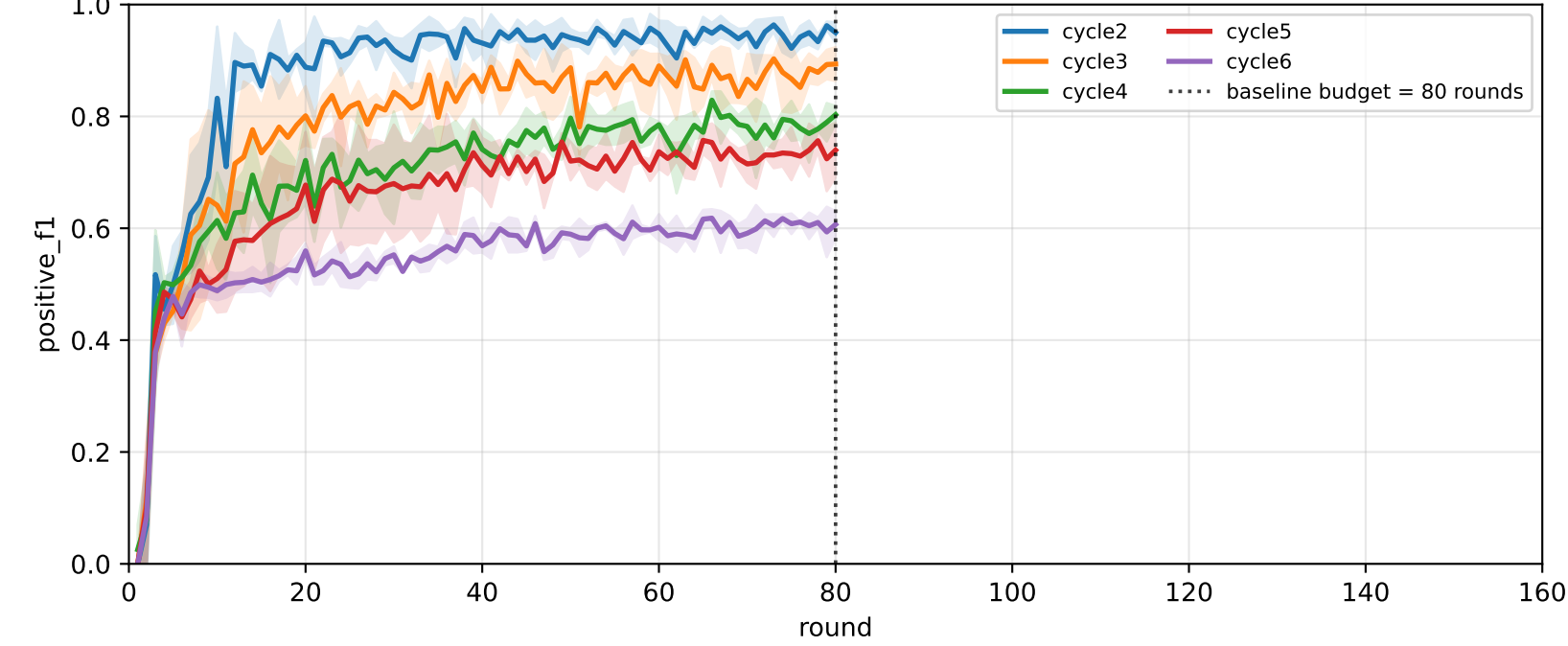
Pooled local centralized positive F1



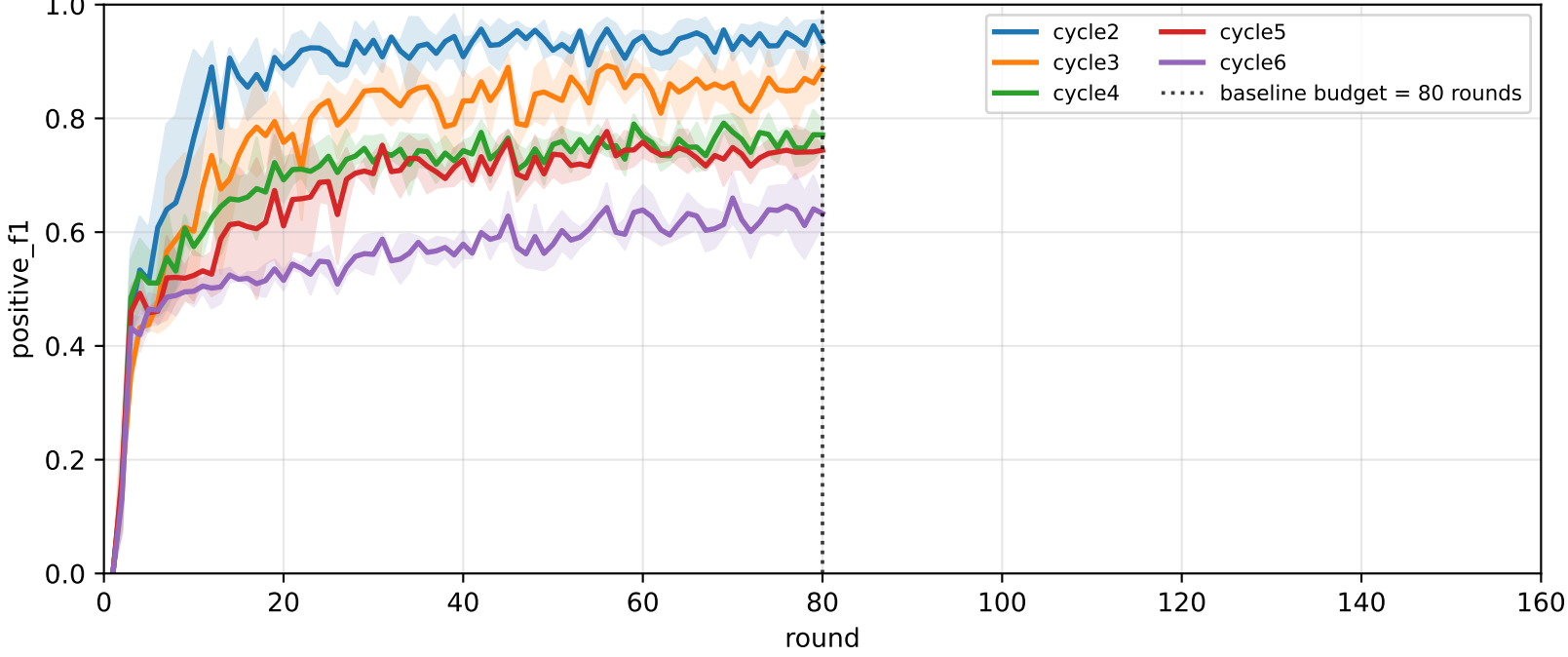
Pooled FedAvg positive F1



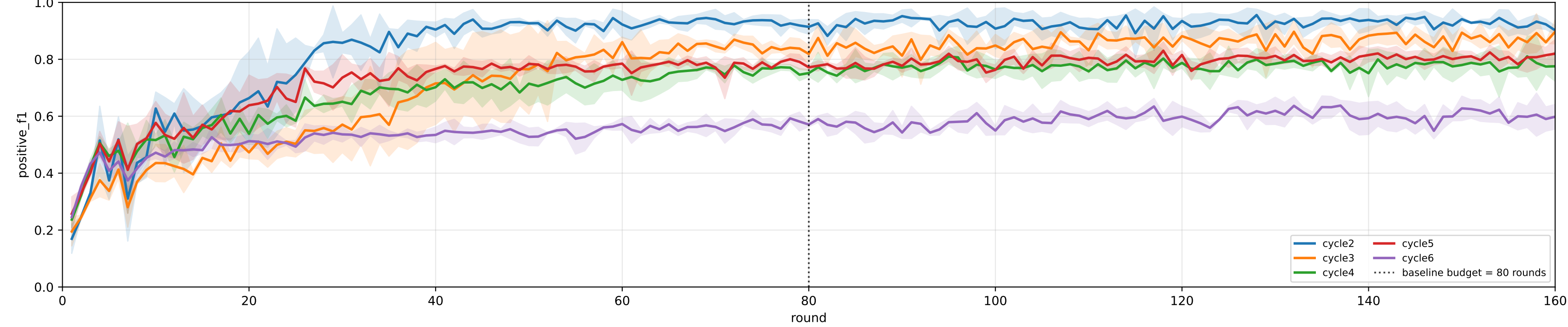
Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)



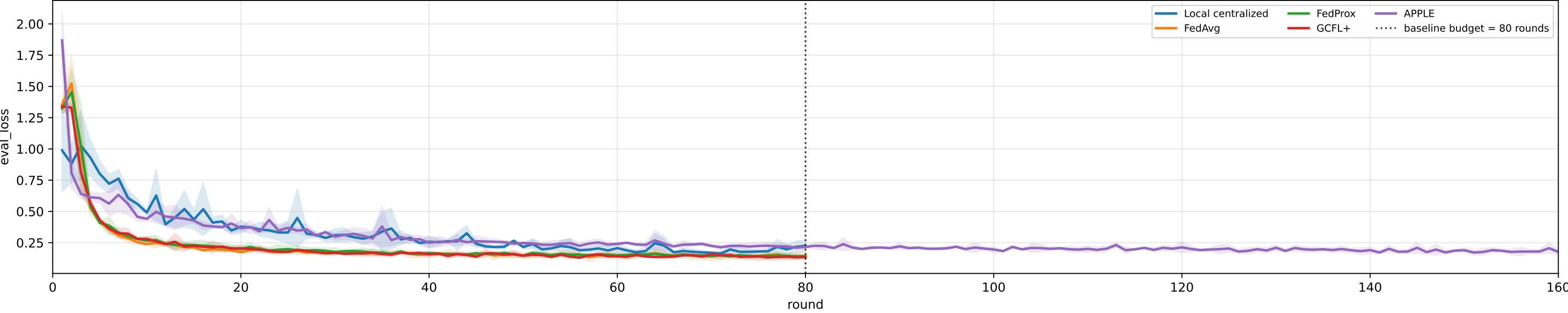
Controlled task-specialization level: High | family aggregated / pooled performance |
family=specialized_cycle6 | n_clients=1 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=specialized_cycle6 | n_clients=1 | cycle2=visible 1.2%±0.0% / true 5.8% | cycle3=visible 1.2%±0.0% / true 6.2% | cycle4=visible 1.3%±0.0% / true 6.6% |
cycle5=visible 1.3%±0.0% / true 6.4% | cycle6=visible 6.3%±0.0% / true 6.3%
specialization=high | specialization_fraction=0.80 | global_visible_support=0.36 | task_jsd_mean=0.162 | task_jsd_median=0.162 | task_jsd_max=0.165 |
family_centroid_jsd_mean=0.162

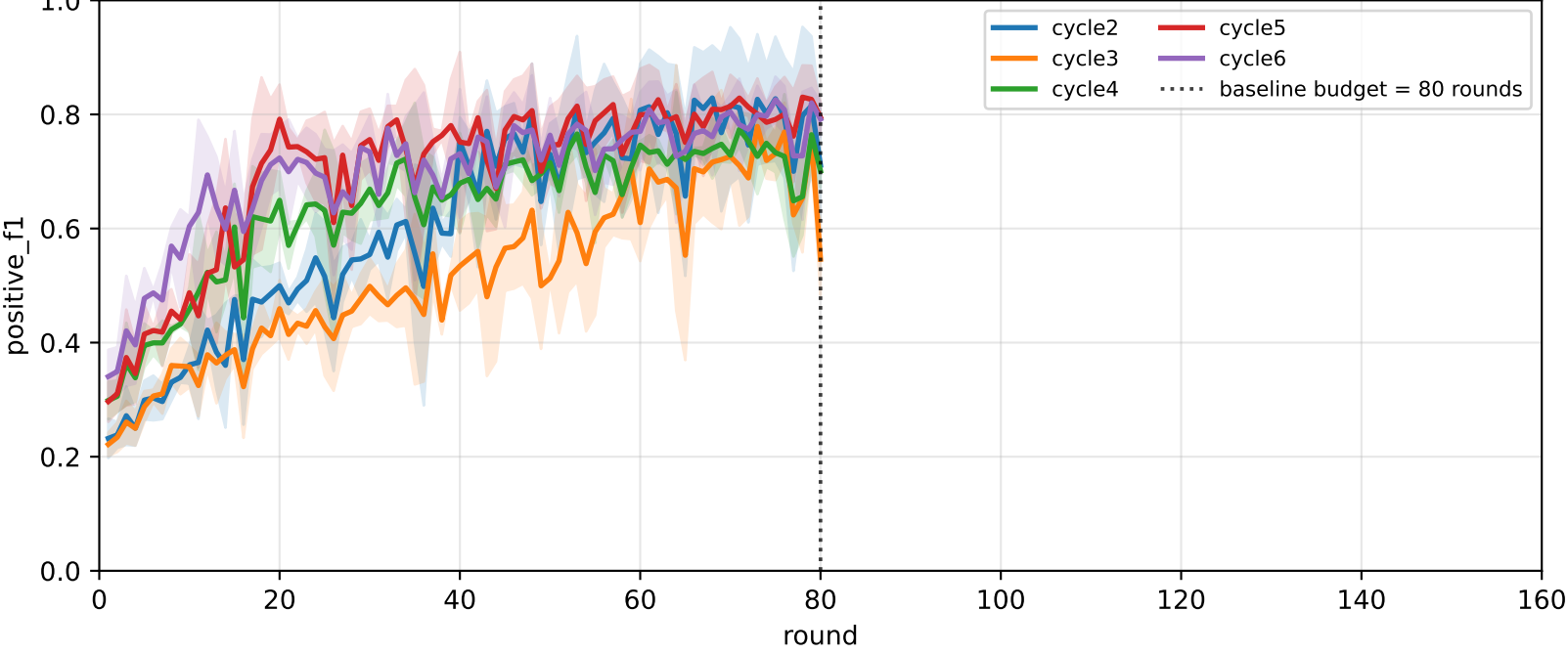
Best logged test performance (baselines 80 rounds; APPLE 160-round long-run ablation)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	78.5% ± 3.5%	79.1% ± 2.8%	83.3% ± 8.5%	79.7% ± 5.3%	76.3% ± 4.2%	77.7% ± 6.0%	78.4% ± 6.9%
FedAvg	77.0% ± 2.2%	80.7% ± 2.4%	95.2% ± 3.0%	92.8% ± 2.9%	81.7% ± 2.4%	75.0% ± 3.6%	58.9% ± 1.0%
FedProx	76.4% ± 1.2%	80.5% ± 0.8%	95.4% ± 0.9%	93.5% ± 2.0%	83.9% ± 1.1%	70.7% ± 3.1%	58.9% ± 1.9%
GCFL+	77.5% ± 1.6%	80.2% ± 1.4%	95.8% ± 1.7%	90.5% ± 1.3%	77.9% ± 1.9%	72.1% ± 0.3%	64.9% ± 3.1%
APPLE	73.8% ± 2.9%	75.2% ± 2.3%	87.7% ± 3.2%	77.0% ± 3.4%	70.2% ± 0.8%	67.2% ± 4.3%	74.2% ± 10.0%

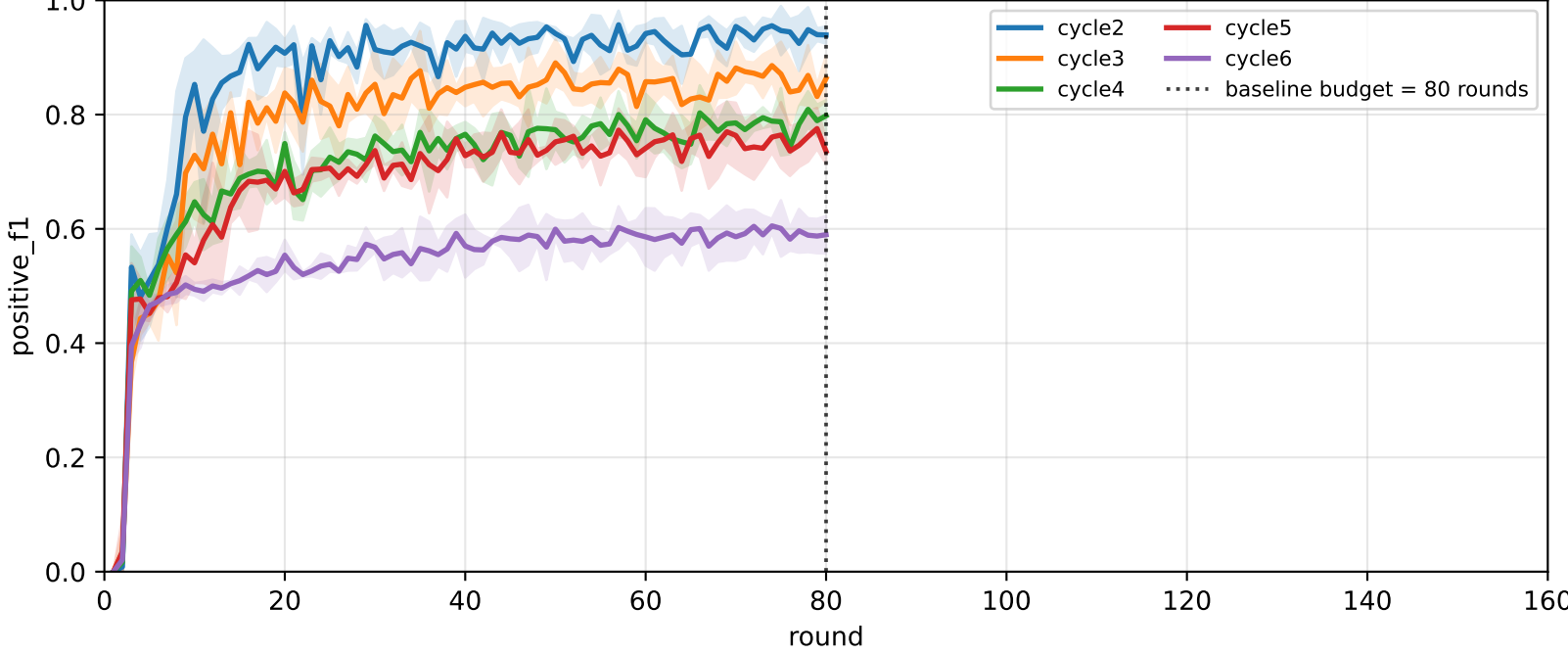
Validation loss comparison (baselines 80 rounds; APPLE 160 rounds)



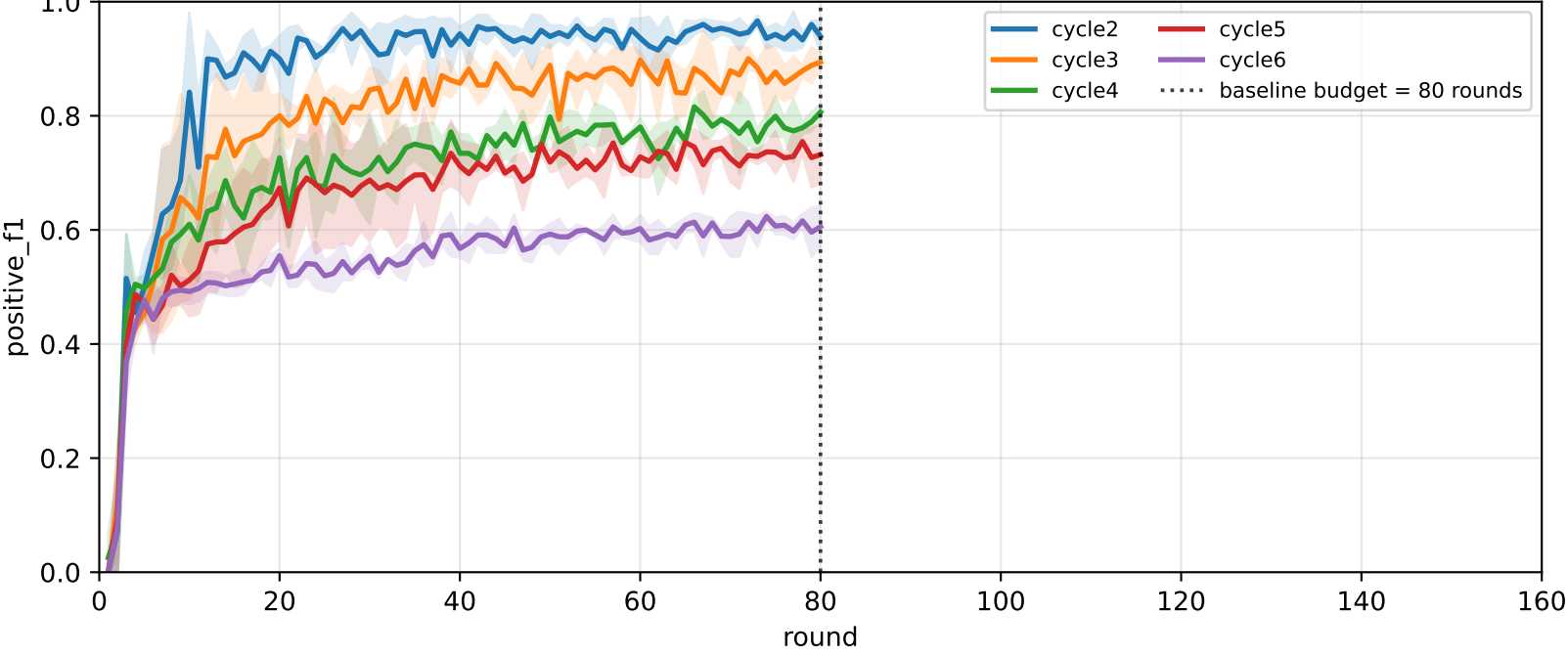
Pooled local centralized positive F1



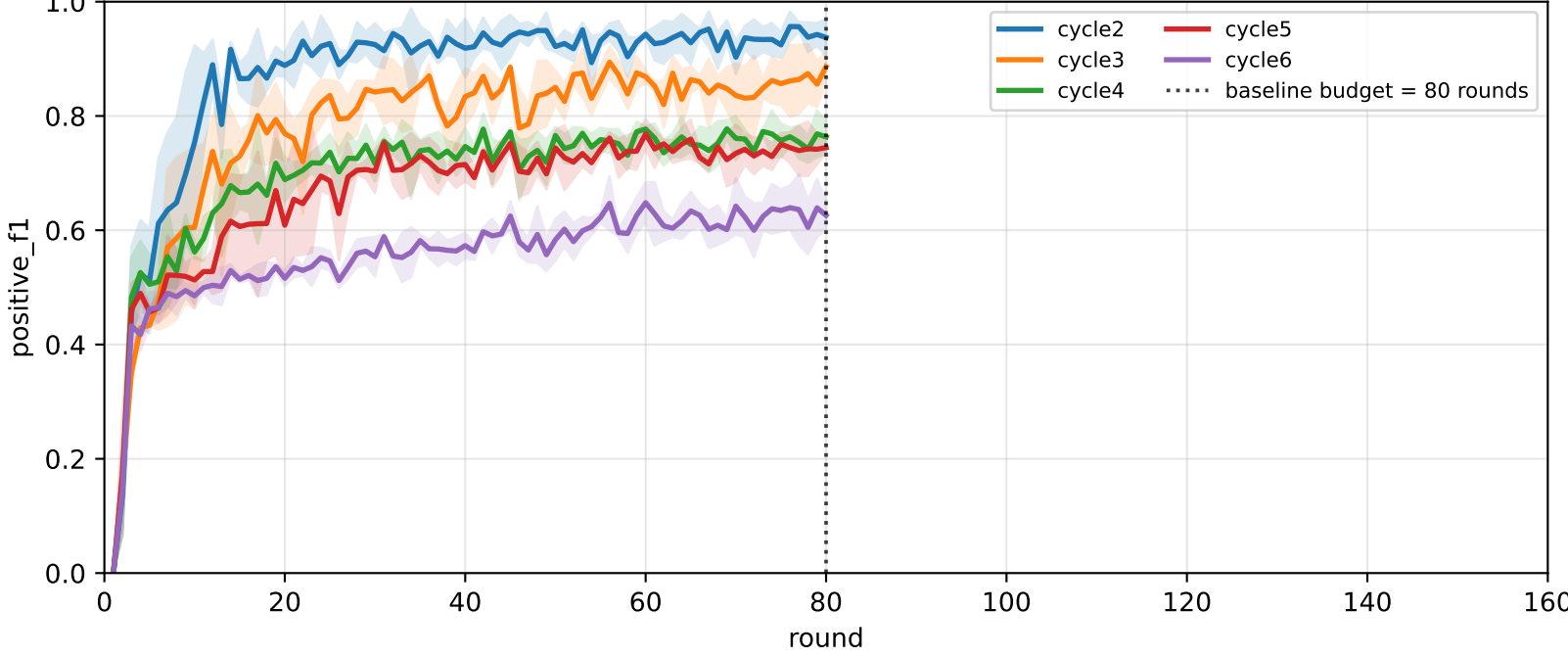
Pooled FedAvg positive F1



Pooled FedProx positive F1



Pooled GCFL+ positive F1



Pooled APPLE positive F1 (after-local selected model, 160-round run)

